

A Theory of Computational Coupling Between Intelligent Systems

Toward a General Foundation for Brain-to-Brain Communication

Ashok Pasala | VIT-AP University | Working Draft v0.2.0 (July 23, 2026)

Status: Working Draft (Version 0.2.0). Formal Core (Sections 3–4) completed; empirical evaluation roadmap and foundational literature canon fully specified.

ABSTRACT

*Efforts toward brain-to-brain communication (BBI) and related brain-computer interface (BCI) paradigms have largely proceeded by designing communication protocols --- fixed encodings, tokenizations, or aligned representational spaces --- without first establishing what quantity such a protocol should maximize, or how to measure whether it has succeeded. We argue this mirrors the state of telecommunications engineering prior to Shannon's definition of channel capacity: a field of designed artifacts without a rigorous target. We introduce a general theory of **computational coupling** between intelligent systems --- biological or artificial --- that defines a substrate-independent, directed, information-theoretic quantity (**coupling capacity**) governing how predictively entangled two systems' internal state trajectories can become, given a bandwidth- and structure-constrained interface between them. We derive three falsifiable predictions from this theory, propose estimators suitable for both simulated multi-agent systems and biological recordings, and outline how brain-to-brain communication, communication protocols, and language itself are best understood as special-case applications of this more general quantity, rather than as the object of study in themselves.*

Introduction

Prior to 1948, telecommunications engineering possessed extensive practical technique but no rigorous mathematical definition of *information* or of a channel's fundamental capacity. Claude Shannon's landmark contribution [shannon1948mathematical] was not a specific telegraph device or radio modem, but a *measurement theory* that specified the theoretical upper bound of error-free transmission over any noisy medium.

- We argue that current efforts in Brain-to-Brain Communication (BBI) and Brain-Computer Interfaces (BCIs) occupy an analogous position today. Substantial engineering ingenuity is devoted to designing tokenization schemes, neural decoders, and aligned embedding spaces, yet the field lacks a governing quantity that specifies:
- What fundamental quantity is being optimized across a brain-to-brain link?
- What are the theoretical upper bounds on information exchange between two dynamic, non-stationary neural manifolds?

The Core Reframe

Communication between two intelligent systems (biological or artificial) is fundamentally not the transmission of a static pre-formed message, but the degree to which their internal state trajectories become **mutually predictive** through a designed or evolved interface map. This reframing subsumes decoding-based and representational-alignment approaches as downstream special cases.

Central Object

Instead of inventing arbitrary brain dictionaries, we introduce **coupling capacity**: a directed, dynamic generalization of Shannon channel capacity, defined over continuous state trajectories rather than static discrete messages.

Scope Statement

This paper presents a measurement theory, not a system proposal. It does not propose a specific BBI hardware architecture or tokenization scheme; it defines the mathematical target against which all such systems must be evaluated.

Related Work

The theory of computational coupling synthesizes literature across several key domains:

- **Information-Theoretic Causality & Transfer Entropy:** Schreiber [schreiber2000measuring] introduced transfer entropy to quantify asymmetric, model-free information transfer between dynamic time series. Massey [massey1990causality] formalized directed information for systems with feedback, laying the foundation for modern measures like directed phase transfer entropy (dPTE). Kraskov et al. [kraskov2004estimating] developed non-parametric k-NN mutual information estimators.
- **Brain-to-Brain Interfaces (BBIs) & Brainets:** Early empirical demonstrations by Pais-Vieira et al. [pais2013brain] (rat-to-rat ICMS) and Rao et al. [rao2014direct] (human EEG-to-TMS motor transfer) proved the physical feasibility of direct brain links. Jiang et al. [jiang2019brainnet] expanded this to multi-person networks (BrainNet). However, these implementations operate under extreme bandwidth constraints (<1 bit/sec) and lack a formal measurement capacity.
- **Cross-Subject Representational Alignment:** Modern state-of-the-art methods utilize Optimal Transport, such as Fused Unbalanced Gromov-Wasserstein (FUGW) by Thual et al. [thual2022aligning], to align functional cortical geometries across subjects.
- **Active Inference & Hermeneutics:** Friston and Frith [friston2015active] modeled communication as mutual free-energy minimization, demonstrating that interacting brains synchronize to minimize joint prediction error under a shared generative model. Hasson et al. [hasson2010speaker] demonstrated speaker-listener neural coupling during natural communication.
- **Emergent Communication in MARL:** Foerster et al. [foerster2016learning] developed Differentiable Inter-Agent Learning (DIAL), demonstrating that deep multi-agent reinforcement learning agents can invent proprietary communication protocols by backpropagating through noisy channels.
- **Large Brain Foundation Models:** Recent advances in self-supervised learning have produced foundational neural codecs, including LaBraM [jiang2024labram] for EEG and BrainLM [caro2024brainlm] for fMRI, establishing universal latent representations of neural state spaces. Scotti et al. [scotti2024mindeye2] introduced MindEye2 for shared-subject visual perception reconstruction.
- **Representation Transfer Threats:** Nakamura et al. [nakamura2024unsupervised] demonstrated zero-shot representation transfer using hyperspherical embeddings. While impressive, their approach performs passive representational mapping; our theory addresses active, closed-loop state trajectory control over bandwidth-limited channels.

Formal Framework

Systems

Let each system $i \in \{1, \dots, N\}$ (biological or artificial) be characterized by:

- A state space M_i (a manifold, possibly high-dimensional), with internal state trajectory $x_i(t) \in M_i$ evolving over a time index t (discrete or continuous).
- A self-predictive model P_i , used by system i to predict its own future state $x_i(t+\Delta)$ from its own past $x_i(\leq t)$.

We deliberately do not assume M_i is shared, aligned, or even comparable in dimension across systems --- a direct departure from hyperalignment-based approaches, which presuppose a common representational geometry.

Interfaces

An interface from system i to system j is a (possibly stochastic) map $g_{\{i \rightarrow j\}} : x_i(\leq t) \mapsto s_{\{i \rightarrow j\}}(t)$, producing a signal $s_{\{i \rightarrow j\}}(t)$ that influences the subsequent evolution of x_j . Interfaces are constrained by a bandwidth or structural budget --- e.g., a discrete channel with vocabulary size K , a continuous channel with bandwidth B , or a hardware-constrained stimulation protocol. Let G denote the admissible set of interfaces under a given constraint budget.

Directed Coupling

We define the directed coupling from i to j , at lag Δ and under interface $g \in G$, as the transfer entropy (directed information): $TE_{\{i \rightarrow j\}}(\Delta; g) = -\log \frac{\mathbb{P}(x_j(t+\Delta) | x_j(\leq t), x_i(\leq t))}{\mathbb{P}(x_j(t+\Delta) | x_j(\leq t))}$, the reduction in uncertainty about j 's future state given i 's state, beyond what j 's own past already provides. This generalizes standard transfer entropy to vector- or manifold-valued states with an explicit lag term, since biological systems exhibit genuine transmission delay and representational drift that a lag-free formulation would misattribute to noise.

Coupling Capacity

DEFINITION: Coupling Capacity

The **coupling capacity** from system i to system j is the supremum of directed coupling over admissible interfaces: $C(i \rightarrow j) = \sup_{g \in G} TE_{\{i \rightarrow j\}}(\Delta; g)$.

This is the direct dynamical analogue of Shannon capacity: where Shannon capacity is a supremum over input distributions of mutual information subject to a power or bandwidth constraint, coupling capacity is a supremum over *interface designs* of directed information between two coupled dynamical systems, subject to an analogous constraint.

Bidirectional Coupling and Asymmetry

Define total coupling as $C_{\text{total}} = C(i \rightarrow j) + C(j \rightarrow i)$, and an asymmetry index $A = \frac{C(i \rightarrow j) - C(j \rightarrow i)}{C(i \rightarrow j) + C(j \rightarrow i)}$; $A \in [-1, 1]$. A is predicted to correlate with task-defined role (e.g., "leading" vs. "following," "speaking" vs. "listening"); see Prediction 3 below.

Falsifiable Predictions

PREDICTION: Capacity--Bandwidth Law

For interfaces parameterized by a scalar bandwidth B , $C(i \rightarrow j; B)$ is monotonically increasing and concave in B , saturating at a value set by $\min(\dim_{\text{eff}}(M_i), \dim_{\text{eff}}(M_j))$ --- the smaller of the two systems' *effective* representational dimensionality --- rather than by the channel's raw capacity. This is directly falsifiable: if measured capacity continues scaling with channel bandwidth well past the point where either system's effective dimensionality should cap it, the theory as stated is wrong.

PREDICTION: Self-Predictive Accuracy Governs Capacity Efficiency

Let R_i denote system i 's self-predictive accuracy --- its own model's log-likelihood in predicting its own future state. Capacity achieved per unit bandwidth, $C(i \rightarrow j)/B$, is monotonically increasing in R_i and R_j jointly: systems with better internal world models extract more coupling from an identical channel.

PREDICTION: Asymmetry Tracks Role

The asymmetry index A correlates with an externally defined task-role variable across domains (multi-agent reinforcement learning, human dialogue, hyperscanning) --- a cross-domain claim, not a redescription of a single-domain result.

Evaluation Metrics and Estimation

Estimating coupling capacity in continuous high-dimensional state spaces requires estimators tailored to the data regime:

Simulated Multi-Agent Systems (Ground Truth): In artificial multi-agent reinforcement learning (MARL) testbeds, full access to internal recurrent states h_t^A, h_t^B permits direct estimation via k -nearest-neighbor (k -NN) Kraskov-St"ogbauer-Grassberger (KSG) transfer entropy estimators. Alternatively, we introduce a *predictive-gain estimator*: $\hat{TE}_{\{A \rightarrow B\}} = L_{\{self\}} - L_{\{joint\}} = E \left[\log \frac{P_{\psi}(x_B(t+1) | x_B(\leq t), x_A(\leq t))}{P_{\phi}(x_B(t+1) | x_B(\leq t))} \right]$ which scales effectively to deep neural networks without suffering from the curse of dimensionality. **Biological Recordings (EEG / fMRI Hyperscanning):** State trajectories are extracted via non-linear manifold learning or pre-trained foundation model encoders (LaBraM, BrainLM) before applying the predictive-gain estimator. Effective Transfer Entropy (ETE) is computed by subtracting surrogate-shuffled baseline estimates to eliminate finite-sample bias.

Falsifiability, Scope, and Discussion

Falsifiability & Scope

A critical requirement for any measurement theory is explicit commitment to falsifiability: **Falsification Criteria:** If measured transfer entropy across an optimized channel scales indefinitely with raw channel bandwidth B past the effective representational dimensionality cap $\min(\dim_{\{eff\}}(M_i), \dim_{\{eff\}}(M_j))$, Prediction 1 is falsified, and the theory as stated is invalid. **Scope Limitation:** This framework quantifies *how much* two dynamic systems become predictively entangled, not *what* semantic content is transmitted. Content-level interpretation is a downstream task dependent on specific representational geometries.

Limitations & Ethical Considerations

Estimator Sample Complexity: High-dimensional transfer entropy estimation remains computationally intensive ($O(N \log N)$ for k -NN estimators). Neural predictive-gain models mitigate this but introduce optimization hyperparameter dependencies. **Non-Stationarity of Neural Manifolds:** Biological neural manifolds exhibit non-stationarity over extended timeframes due to plasticity, requiring adaptive windowing in transfer entropy estimators. **Ethical Considerations for Stimulation Work:** Eventual bidirectional closed-loop BBI prototypes involving non-invasive TMS or invasive microstimulation introduce ethical questions regarding agency, cognitive privacy, and involuntary mental state alignment. **Comparison with Integrated Information Theory (Φ):** While Giulio Tononi's IIT quantifies integrated information *within* a single system, Coupling Capacity $C_{\{couple\}}$ quantifies directed information flow *between* dynamic systems across a bandwidth-limited interface.

Multi-Year Empirical Roadmap

- **Paper 1 (Multi-Agent RL):** Validate Predictions 1 and 2 in controlled cooperative games (PettingZoo simple_speaker_listener_v4), sweeping channel quantization $B \in [1, 2, 4, 8, 16, 32]$ bits/step.
- **Paper 2 (Hyperscanning):** Test Predictions 1--3 against public human dyadic EEG corpora (OpenNeuro ds007764 DUET, ds007471 Joint Agency EEG) using FUGW functional alignment.
- **Paper 3 (Causal Manipulation):** Causal non-invasive intervention manipulating feedback latency and bandwidth in human dyads.
- **Paper 4 (Learned BCP Prototype):** Use C_{couple} as an explicit differentiable loss function to train a closed-loop BBI prototype.

Conclusion

We have introduced a general Theory of Computational Coupling between intelligent systems, reframing Brain-to-Brain Communication (BBI), communication protocols, and language from designed engineering artifacts into testable applications of a single underlying quantity: **Coupling Capacity** (C_{couple}). By generalizing Shannon's channel capacity to dynamic, self-predictive state trajectories on neural manifolds, this framework provides the field with a rigorous measurement target.

References & Literature Database

All citations referenced in this working paper (Shannon 1948, Schreiber 2000, Pais-Vieira 2013, Rao 2014, Foerster 2016, Thual 2022, LaBraM 2024, BrainLM 2024, MindEye2 2024, Nakamura 2024) are formally verified and cross-referenced in `paper/references.bib` and `literature/literature_review.md`.